Name: ______________________

Worksheet

Arithmetic sequences

Task A: Identifying arithmetic sequences

1) For each set of terms, work out if they could form an **arithmetic sequence** or not. Explain your answer.

a) $\boxed{5}$, $\boxed{9}$, $\boxed{}$, $\boxed{17}$, $\boxed{}$, $\boxed{25}$, ...

b) $\boxed{0.1}$, $\boxed{}$, $\boxed{}$, $\boxed{}$, $\boxed{0.3}$, $\boxed{}$, ...

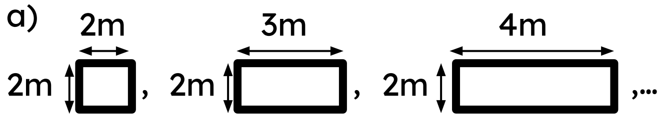
c) $\boxed{4}$, $\boxed{}$, $\boxed{16}$, $\boxed{}$, $\boxed{}$, $\boxed{28}$, ...

d) $\boxed{8}$, $\boxed{}$, $\boxed{}$, $\boxed{}$, $\boxed{}$, $\boxed{-2}$, ...

e) $\boxed{2\frac{1}{2}}$, $\boxed{}$, $\boxed{}$, $\boxed{2}$, $\boxed{}$, $\boxed{\frac{5}{3}}$, ...

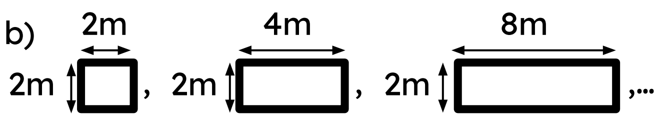
2) For each sequence work out the perimeter and area of each term and state whether they form a **linear sequence** or not.

a)



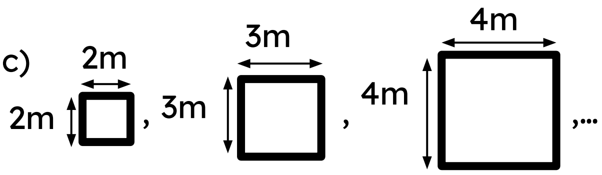
Term	1	2	3
Perimeter			
Area			

b)



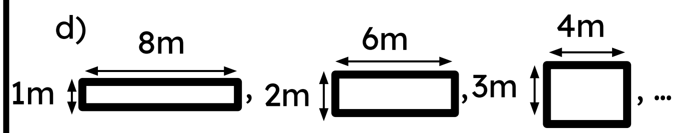
Term	1	2	3
Perimeter			
Area			

c)



Term	1	2	3
Perimeter			
Area			

d)



Term	1	2	3
Perimeter			
Area			

Task B: Describing arithmetic sequences

1) For each sequence state whether it is **arithmetic** or not. If it is **arithmetic**, describe the sequence.

a) 6, 15, 24, 33, ...

b) 5, 12, 15, 22, ...

c) 5, −3, −11, −19, ...

d) 1, 3, 9, 27, ...

e) 1.5, 2.6, 3.7, 5.8,...

f) 25, 15, 10, 5, ...

g) $x,\ x+2,\ x+4,\ x+6$ ...

h) $2x,\ 4x,\ 8x,\ 16x$ ...

i) $x,\ 2x+1,\ 3x+2,\ 4x+3$ ...

j) $x-2,\ x,\ x+2,\ x+4$ ...

2) Below are the first four terms in four different **linear sequences**: Fill in the table with true or false for each sequence.

Sequence	1, 5, 9, 13,..	7, 13, 19, 25,..	80, 75, 70, 65,..	22, 19, 16, 13,..
Contains negative numbers				
Common difference is positive				
Has 1 as a term				
Has 25 as a term				
Has values greater than 50				